1

Chapter 1 – Introduction to open source development

Open source software development is a methodology for creating software products, from the design and development to distribution. Under this methodology the author offers access to the source code. This chapter gets you started into this fascinating software development world. It teaches you how it all started, and what the direction will be for the coming years.

In this chapter you will learn about:

- A brief history about open source software development
- The evolution of the open source movement
- Open source versus free software
- Advantages and disadvantages of open source
- Trends and perspectives
- Career prospective in the open source world

1.1 A brief history about open source development

The story of open source development started long before Richard Stallman created the Free Software movement. In the 50's and 60's almost all the software that existed was mostly produced by research institutes. Software was not seen as a product. In those times, computer companies were in the hardware business; and software was made freely available to encourage hardware sales. The source code was distributed with the software because users often had to change the code to fix bugs or add new features to support hardware issues. During this time software was developed and distributed by communities of user groups and no effort was needed to make it freely available.

Things started to change in the early 1970's when operating systems and compilers began to grow very fast, with the emergence of micro-processors. For almost a decade until the early 1980's computer vendors and software companies began to routinely charge for software licenses, and sell software as a product imposing legal restrictions on new software developments through copyrights, trademarks, and leasing contracts. At that time

two different groups, both in United States, were establishing the roots of the current open source software (OSS) philosophy.

In the East coast, a programmer at the MIT Artificial Intelligence Lab launched the GNU Project and the Free Software Foundation, his name was Richard Stallman. On the West coast, the Computer Science Research Group (CSRG) of the University of California at Berkeley were improving the UNIX system, and started to build lots of applications which quickly became known as “BSD UNIX”. With the advent of Usenet, an Internet user group, programming communities started to share their software and contribute to each others' work. This is when the real development of the open source movement began.

1.2 The evolution of the open source movement

We start recording the evolution of open source development from the creation in 1986 of the Free Software Foundation by Richard Stallman (who likes to use his initials RMS). After this foundation was established, several major open source projects were initiated as shown in *Figure 1.1*.

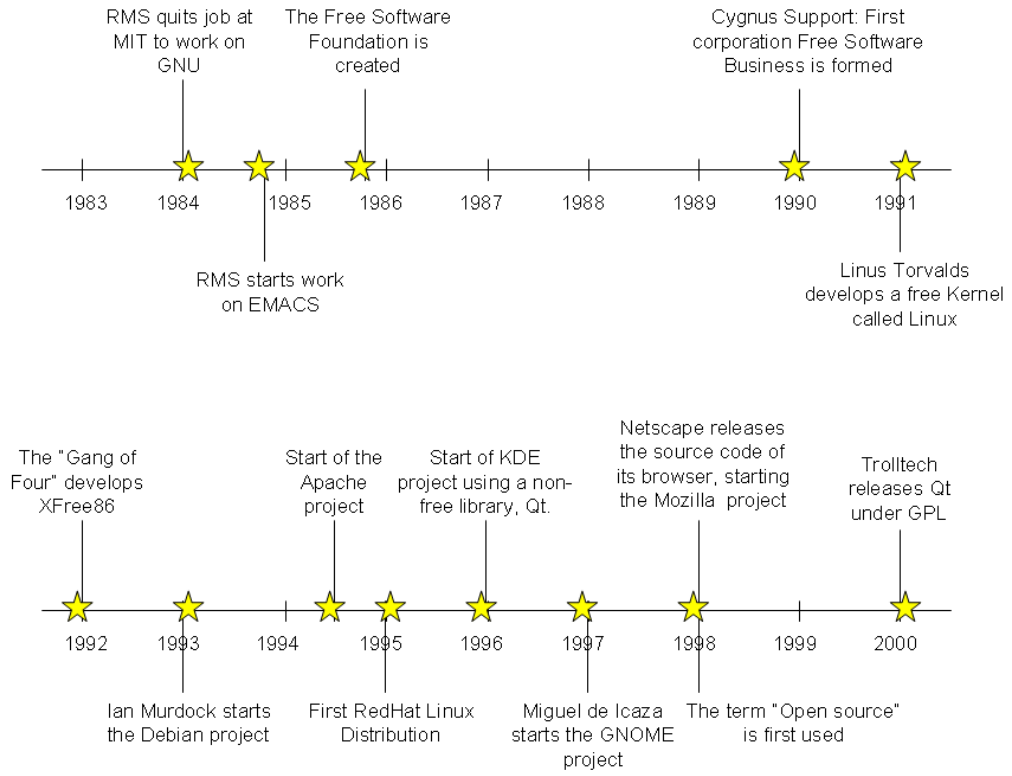


Figure 1.2 - Evolution of Open Source development

In 1984, Stallman started the GNU project with the ultimate goal of building a free operating system. Within this project, he wrote the EMACS editor, which is considered by many as the most powerful text editor.

In 1985 Stallman created the Free Software Foundation (FSF), a nonprofit organization dedicated to the elimination of restrictions on copying, redistribution, understanding and modification of computer programs. The organization's goal was to promote the development and use of open source software in all areas of computing, particularly in helping to develop the GNU operating system and its tools. Until 1990 the foundation was dedicated to writing more software. Today, it coordinates many independent open source projects, where the FSF's focus has shifted to the structural and legal aspects of the open source community. As a legal tool, the GNU General Public License (GPL) was designed not only to ensure that the software produced by GNU would remain free, but to promote the production of more and more open source software.

In 1987 Stallman created an open source compiler, the GNU C Compiler (GCC) with the idea to encourage more open source code contributions. Nowadays the GCC compiler is often chosen as the favorite to develop software that needs to be implemented in various types of hardware and supports C, C++, FORTRAN, Ada, Java™, Objective-C and Pascal.

For many years, employees of Cygnus Support, a small company founded by Michael Tiemann, David Vinayak Wallace, and John Gilmore were the maintainers of several key GNU software products, including the GNU Debugger (gdb) and GNU Binutils. This company was also one of the major contributors to the GCC project.

In 1991 a student of the Department of Computer Science at the University of Helsinki in Finland, Linus Torvalds, created with the help of several volunteer programmers through Usenet the Linux® kernel. Some time after, the kernel grew with the assistance of developers around the world as open source software. A few years later a full open source operating system was available and released as GNU/Linux.

In 1993 the Debian GNU/Linux operating system was created by Ian Murdock in order to assemble all the GNU tools that existed at the time and the Linux kernel. The Debian Project grew slowly and gained notoriety when the program dpkg was released. This program is the basis of the Debian package management system and is used to install, remove, and provide information on the Debian software packages (.deb).

One of the most important events in open source development happened in 1994 when Robert McCool developed the Apache HTTP server. This Web server played a key role in the development of the World Wide Web, and it was the first open source alternative to the Netscape Web server. Today, more than 100 million Web sites use Apache as their Web server of choice, as shown in *Figure 1.2*.

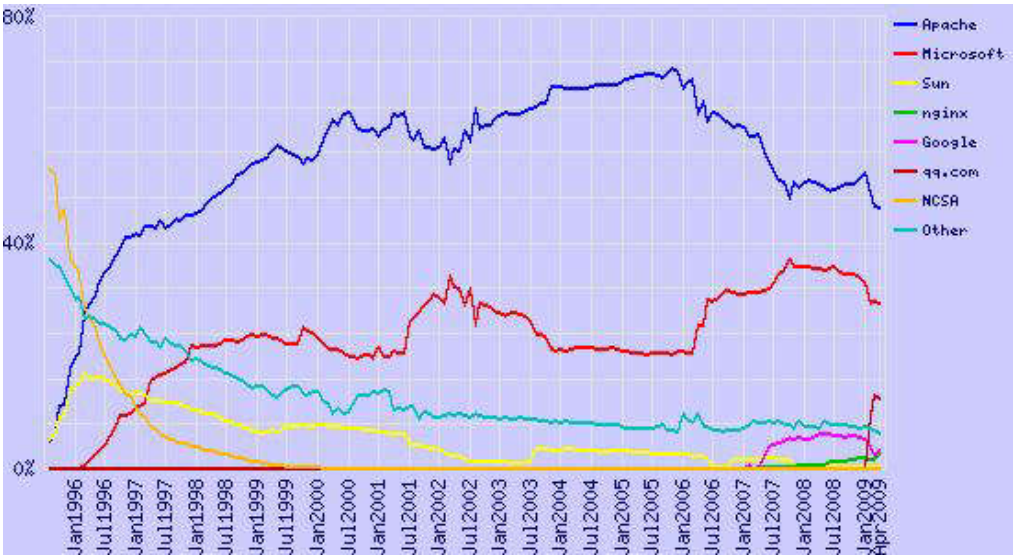


Figure 1.2 – Market share for top Web servers across all domains according to Netcraft (http://www.netcraft.com)

In 1995 Marc Ewing created his own Linux distribution called RedHat. RedHat is nowadays a major company that provides operating-system platforms along with middleware, applications, and management products, as well as support, training, and consulting services.

In 1996 the KDE and GNOME desktop environments were developed, providing basic desktop functionality for daily needs as well as development tools.

Two years later, in 1998, many software companies started to accept the open source movement when Netscape Communicator source code was made open source, and the Mozilla Foundation was established. It is in that year as well, that the term ***open source*** was first used at a conference in California. This term was created by Eric Raymond and Bruce Perens to promote the free software philosophy in the corporate world.

1.3 FLOSS - Free, *libre*, open source software

Until now the use of the term "open source" has many opponents. From one side Stallman and others with similar thoughts object to "open source" as they say it does not make users realize the freedom that the software in question gives to them And when Stallman uses the term "free software", he is not referring to price as indicated by his quote, "*Free software is a matter of liberty, not price. To understand the concept, you should think of 'free' as in free speech, not as in 'free beer'*". On the other hand, using the term "free software" was commonly confused with the no-charge connotation, which obviously made business organizations uneasy.

In order to end this discussion other terms have been proposed, one of them is FLOSS – Free, *libre*, open source software, or Free (as in libre) open source software. There are

three terms to explain this concept. As already mentioned, **free software** is a philosophical concept that aims to convey the idea of software that can be used, studied and modified without any restriction. It does not refer to price, but liberty. This software can be freely distributed. Because many people can confuse the concept of free software with freeware (closed-source software freely distributed by the author) the concept of Libre Software was created. **Libre** is a word used by several Romanic Languages such as Spanish, Portuguese and French. This word does not exist in English, but what it means is free as in "free speech", freedom as a right that a citizen has and not freedom associated with zero cost (Gratis). **Open source** on the other hand is a movement which emphasizes on source code of software being freely available. It also talks about certain criteria which must exist in the license of software to make it open source.

In this book when we used the term ***open source***, we mean FLOSS. In essence the freedom here refers to what an individual developer or user has with the code. It does not refer to a no-charge license. We discuss more details on the various connotations this phrase has in the open source world in the chapters to come.

Did you know?

The first use of the phrase “free open source software” on Usenet was in a posting on 18 March 1998, just a month after the term “open source” itself was introduced.

Did you know?

The IBM® DB2 Express-C database is free as in free beer. Check out these Web sites for more details:

http://freedb2.com/2009/08/06/db2-express-c-those-who-use-oracle-like-it-a-lot/

http://db2express.com

http://tldp.org/HOWTO/DB2-HOWTO/whyexpc9.html

1.4 Advantages and disadvantages of open source

Based on the fact that open source software is free and can be seen by everyone has great advantages. But the fact that it is being developed by a nonprofit community has some disadvantages. In this section we discuss the pros and cons of open source software maintenance and development.

1.4.1 Pros

Open source software has lower monetary costs as development, support and license costs are fairly minimal when compared to proprietary software. This does tempt many organizations to use open source software in their business model. In fact many companies do business with open source, IBM being one of them. For example, IBM offers a quality service with the Linux operating system. Another company, RedHat is selling the Linux operating system with support services.

As far as security and reliability goes, open source is a great model because, as many people analyze the source code, a safer and more secure code will be produced. Since open source software has so many people participating in its development, some programs are faster and scale better than the proprietary counterparts. Also depending on usage and need, open source software is also changed by experienced users thereby making code more stable.

Open source is also the answer to the incompatible formats in proprietary software, because it only uses open standards, that is, standards that are known or are accessible to all the people. One such example is OpenDocument Text (.odt), which is an open standard for word processor documents.

From a corporate perspective, companies that use open source do not have to worry about complicated licensing, and thus, do not suffer the risk of having illegal copies that infringe copyrights. Therefore, they don't need anti-piracy measures, such as CD keys, product activation and serial keys.

Open Source software is community driven and community serving; a large number of bright, and generous developers work openly and with the whole community. For example when an open source program crashes it provides useful information to find the source of the error or to report a possible bug.

Open source software is independent of companies and its main authors. If the company goes bankrupt or the authors fail to maintain the program, the code continues to belong to the community. Therefore, many people believe open source software can live by itself. As long as there are passionate contributors from the community, this is indeed the case.

1.4.2 Cons

Open source software has been focused to provide solutions to servers rather than to desktop computers. As a result, adoption in the desktop arena is much slower. For example, Linux desktops are still not used as much as Microsoft® Windows®. In addition, many software is not yet compatible with open source. When a user chooses a Linux desktop he has to remove several software that are not supported on Linux, or in some cases, there is no similar nor viable open source application. A good example is the gaming industry, which is still very focused on Windows.

Excluding companies that sell open source combined with technical support; proprietary software offers better service and support. The quality and availability of assistance in an open source project is proportional to the interest and use of the program by the community. An open source tool with few users can be poorly documented and have almost no means to help you understand it.

1.5 Open source trends and perspectives

The advantages of OSS outweigh its disadvantages, this is why companies are starting to pay close attention to open source. Increasingly, many companies are using open source software tools for development and test; but open source is quickly gaining market share

as an alternative in production environments. A well known example is the LAMP stack – LAMP being Linux, Apache, MySQL® and PHP/Perl/Python. It is arguably the widest adopted software stack for the deployment of Web applications.

Many other open source projects that emerged since the 80's are still going strong, with many users and better product quality. This is the case of the Eclipse IDE (Integrated Development Environment). Eclipse was developed by IBM in 2001, and donated to the open source community in 2004. Today, Eclipse is one of the widest IDEs in use worldwide.

There are more and more companies whose business model relies on open source. For example, some companies specialize in delivering an integrated and fully tested environment consisting of different open source software like LAMP. The fact that such companies exist is the greatest measure of the success of open source community developed software.

Chapter 2 discusses in more detail other business models used in the open source world, and more details about Eclipse too. Other chapters in this book will discuss other successful open source projects in more detail.

1.6 Career path

We don't want to undermine your intelligence by making you believe that the number of jobs in OSS is much higher than that for proprietary software. However, though smaller, it is a niche market.

Today it would be easier to get a C/C++ programmer job. In order to get developers who have programmed in newer technologies like Python, or Ruby, and who have understood the community, its trends, and its ways of working would certainly take a lot more time and effort. Just as we make correct technology choices for a software solution based on its architecture, organizations need to make choices to find the right fit for the right job. With the growing trends of involvement and usage of OSS in companies the demand for open source developers to take up this work is definitely on the rise. IBM itself has shown a lot of involvement in open source software like Eclipse, Linux, and so on.

In summary, OSS jobs are increasing. It’s still a niche area, but its growth potential makes it worth exploring. Needless to say that being an expert in a niche area always calls for a premium salary for oneself.

1.7 Exercises

Review the following link to learn about the involvement of IBM in OSS development.

http://www.accessmylibrary.com/article-1G1-133183812/history-ibm-open-source.html

1.8 Summary

In this chapter you learned about the history of open source and how it evolved over the years. You also learned about the underlying difference between the terms "free software"

and "open source software". This moved us to discussing some of the advantages and disadvantages of open source software and how working in the open source space could benefit your career in the long term.

1.9 Review questions

1. When was the term "Open source" first used and by who?
2. What does FLOSS stands for?
3. List the pros of FLOSS
4. List the cons of FLOSS
5. Which foundation was started when Netscape Communicator software was made open source?
6. Which of the following open-source projects appears in the 90s?
   A. EMACS
   B. XFree86
   C. KDE
   D. Eclipse IDE
   E. None of the above
7. Who was the founder of the Free Software Foundation?
   A. Richard Stallman
   B. Ian Murdock
   C. Miguel de Icaza
   D. Linus Torvalds
   E. None of the above
8. One of the two groups who established the roots of OSS were:
   A. The founders of the GNU project
   B. The creators of BSD UNIX
   C. The founders of the GNOME project
   D. A & B
   E. None of the above
9. In Stallman's definition of Free Software, "Free" means:
   A. Free as in "Free beer"
   B. Free as in "Free speech"

C. Free as in "Free of charge"

D. All of the above

E. None of the above

10. In Stallman's definition, Free Software is:

A. Freeware

B. Libre

C. Software that can be modified, studied and distributed with no restriction, but may not be free of charge

D. B and C

E. None of the above